Protein folding as a configurational reconstruction

Bio-epistemic application of the General Epistemic Dispersion Theory to adult human hemoglobin

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Note of purpose

This work proposes a theoretical application of General Epistemic Dispersion Theory to protein folding, using adult human hemoglobin (HbA) as a case study. It does not introduce a new biochemical folding mechanism and does not present new experimental biological data. The objective is to verify whether the correlation between reusable structural information, residual configurational uncertainty, and additional dynamical information can be formulated consistently in the domain of molecular biology.

Abstract

General Epistemic Dispersion Theory distinguishes between configurational information implicitly embedded in a reusable structure and additional information that must be explicitly transmitted to determine a global configuration. In this work, we propose an application of this formalism to protein folding, with particular reference to adult human hemoglobin (HbA).

The starting point is that the primary sequence does not encode an explicit list of three-dimensional coordinates of the finished protein. Instead, the configuration emerges from the interaction between residue identities and orders, the physicochemical environment, the energetic landscape, cofactors, and, when present, cellular support mechanisms. We therefore introduce a conditional decomposition of configurational uncertainty: $H(\Pi \mid F) = I(\Pi ; S \mid F) + H(\Pi \mid F, S)$, where S represents the primary structural core, H_{res}^{bio} the residual uncertainty on the final configuration.

In the quasi-identifying limit H_{res}^{bio} can be small; when more states remain accessible, it remains positive. Chaperones [4, 5] and other dynamical contributions can be represented, depending on the model boundary, as part of the space of feasible configurations $\mathcal{P}$ or as additional information M_{Π}. The case of HbA [8, 9], including the role of AHSP and the examples of HbS [11] and Hb G-Copenhagen [12], shows how single variations can have very different configurational effects. The analysis does not pretend to directly quantify biological entropy, but proposes a falsifiable and operationalizable informational framework for further studies.

Keywords: epistemic dispersion; protein folding; structural information; conditional entropy; hemoglobin; chaperone; AHSP; protein configuration; the general epistemic dispersion theory.

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1. Introduction

One of the central problems of molecular biology is understanding how a linear sequence of amino acids can give rise to a functional three-dimensional structure. The classical formulation associated with the thermodynamic hypothesis of Anfinsen [3] states that, under appropriate environmental conditions, the native conformation is determined by the totality of interatomic interactions associated with the amino acid sequence and the physical-chemical environment. Modern folding theory has then shown that this relationship should not be interpreted as a random combinatorial search nor as a universally autonomous process: energetic landscapes, kinetics, cellular crowding, chaperones, cofactors and molecular partners can contribute in an essential way to the formation or stabilization of the functional state.

From an informational perspective, however, a simple yet relevant distinction remains: the genome and the mRNAs encoding globins do not contain an explicit list of the spatial coordinates of each atom of the adult protein. The cell does not transmit, with each synthesis event, a complete three-dimensional manual of the HbA tetramer. A substantial portion of the information required for configuration is instead operationalized by the primary structure and reusable physicochemical constraints.

General Epistemic Dispersion Theory (GEDT [1]) has been proposed to describe, in general terms, systems in which a global configuration can be determined without all the relational information having to be re-explained at each execution. In this framework, information does not disappear: it is partitioned between a reusable structure (a structural core S) and a dynamic residue that must be resolved during execution (through the explicit provision of additional metadata).
This article applies this distinction to protein folding. Adult human hemoglobin is used as a case study because it combines a well-characterized primary structure, multimeric assembly, heme incorporation, sensitivity to specific amino acid substitutions, and the participation of an erythroid-specific chaperone, AHSP [6, 7]. The goal is not to reinterpret the entire biochemistry of folding through a single equation, but to test whether epistemic formalism can coherently describe the informational limits of the specific configuration in the space of possible configurations.

2. Purpose, terminology and nature of the proposal

The term "epistemic" is used here in an informational sense [1, 2] and not an anthropomorphic one. Molecules and cells are not treated as cognitive agents. The object of analysis is the amount of information required, within a model, to determine a global configuration from local components, structure, and execution conditions.
Similarly, the term "metadata" does not indicate a biochemical category. In the language of GEDT, it designates control, relational, or configuration information that does not coincide

with the local payload under consideration. In the biological domain, this terminology must therefore be understood as an abstract mapping, not as a replacement for molecular nomenclature.
The proposal is deliberately conditional. It does not assume that all proteins have a single native state, that folding is always spontaneous, or that the amino acid sequence in the primary structure alone is sufficient in any environment. The environment, cofactors, chaperones, and post-translational modifications can reduce or transform configurational uncertainty and must be considered in model development.

3. Theoretical framework: from GEDT to the biological domain

In the general formalism, let X be the global raw state of a started instance, F = D(X) the set of fragments generated by X, let $\mathcal{P}$ be the finite space of possible feasible configurations, Π a random variable defined in $\mathcal{P}$ representing the realized global configuration of the fragmented instance, including instance-specific roles, ordering, relations, and associations; and let S be a reusable structural core containing relations, roles, or constraints capable of reducing uncertainty about the configuration.
The theory distinguishes the information absorbed by the structure from the residual portion that remains to be resolved.

$$H(\Pi \mid F) = I(\Pi ; S \mid F) + H(\Pi \mid F, S) \quad (1)$$

The space of feasible configurations $\mathcal{P}$ incorporates the physical and biochemical conditions within which the structural core S is instantiated. In the biological application, these conditions include the relevant physicochemical constraints, such as the aqueous environment, temperature, pH, ionic conditions, molecular interactions, and other reusable constraints that define the accessible configurational landscape.
These conditions do not require the introduction of an additional conditioning variable in the core epistemic decomposition.
We therefore define the biological epistemic dispersion and the residual configurational uncertainty as:

$$D_{epi}^{bio}(S ; \Pi \mid F) := I(\Pi ; S \mid F) \quad (2)$$

$$H_{res}^{bio} := H(\Pi \mid F, S) \quad (3)$$

The term D_{epi}^{bio} measures how much uncertainty about the configuration Π is absorbed by knowledge of the structural core S, once F has been fixed and the space of feasible configurations $\mathcal{P}$ has been defined. H_{res}^{bio} instead represents what remains undetermined after specifying the components and structure.
GEDT Theorem 1 provides a lower bound on the average length of any additional configurational transcript M_Π required to solve the residue, up to a correction associated with the admitted error:

$$\mathbb{E}[\ell(M_\Pi)] \geq \max\{0, H_{res}^{bio} - \delta_{\mathcal{P}}(\varepsilon_\Pi)\} \quad (4)$$

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It is essential not to misunderstand the meaning of the bound: a lower bound equal to zero does not prove that the real transcript is zero.
In the ideal Identifying Regime, if the configuration Π is determined by F and the structural kernel S within the feasible space $\mathcal{P}$, that is, if there exists a deterministic function Π = g(F, S), then $H_{res}^{bio} = H(\Pi \mid F, S) = 0$. In this case, the bound of Theorem 1 does not impose any positive dynamic informational cost, and a reconstruction with $M_{\Pi} = \emptyset$ can exist.
In real biological systems, $H_{res}^{bio} \geq 0$ and can be small without necessarily being zero.

4. Bio-epistemic mapping

To make the mapping explicit, we adopt the following model boundary. The choice of boundary is important: a constitutively available chaperone can be included in the $\mathcal{P}$ space, while a dynamic and specific intervention can be represented in M_{Π}. The two descriptions are not contradictory as long as the model declares which information is treated as reusable background and which as additional execution-specific contribution.

Variable	Biological interpretation	Observation
F	indexed collection of individual amino acid residues, each with its own identity, considered independently of their ordering and overall three-dimensional configuration	Local system components.
S	Ordered primary sequence and covalent connectivity of the polypeptide.	Structural core that restricts the configurational space.
$\mathcal{P}$	Physically accessible configuration space/landscape, which also incorporates the effects of the execution environment on individual configurations: solvent, pH, temperature, ionic strength, crowding, cofactors, and cellular factors included as background.	Energy and geometric feasibility constraints.
Π	Final functional configuration or conformational ensemble of interest.	Global output of the configurational reconstruction.
M_{Π}	Additional dynamic information or assistance	It may include, depending on the model boundary,

	not already absorbed in S or incorporated into the definition of the finite space $\mathcal{P}$.	chaperones, targeting, cofactor delivery, or other specific interventions.

In this mapping, $\mathcal{P}$ does not represent the complete microscopic continuum of all possible atomic coordinates, but a finite representation of the configurations or conformational classes distinguishable at the resolution adopted by the model. Microscopically different configurations can therefore belong to the same configurational class in $\mathcal{P}$.

5. Adult human hemoglobin as a case study

5.1 Primary structure and assembly of HbA

The primary human adult hemoglobin, HbA, is an $\alpha_2\beta_2$ heterotetramer. The canonical UniProt sequences of the human subunits comprise 142 residues for α-globin and 147 for β-globin, respectively, including the initial methionine; in the historical numbering of the mature chains, 141 and 146 residues after removal of the initiator are commonly considered. Each subunit binds a heme group, with iron in the Fe^{2+} state in the functional form capable of binding oxygen.

This architecture is useful for the present model because the primary sequence does not contain explicit three-dimensional coordinates of the tetramer. Achieving the functional state depends on hydrophobic interactions, hydrogen bonds, electrostatic interactions, heme coordination, and compatibility of the interfaces between subunits. These relationships are reusable: they do not need to be rewritten as a list of coordinates for each new synthesis.

At the same time, HbA demonstrates why it would be incorrect to equate the entire process with autonomous folding of the sequence alone. Hemoglobin biosynthesis requires globin production, heme synthesis and incorporation, correct subunit stoichiometry, and control of the stability of the free chains.

5.2 Explicit information in globin transcripts [10]

The α and β subunits do not derive from a single "mRNA-HbA," but from distinct transcripts of the α-globin genes (HBA_1/HBA_2) and the HBB gene.

Mature transcripts comprise a coding region and regulatory components such as the 5′ cap, 5′ UTR, 3′ UTR, and poly(A) tail.

These elements are biologically necessary for stability, export, recruitment of the ribosomal translation machinery, and expression regulation.

In epistemic mapping, these elements can be considered explicit control information. However, they do not constitute a three-dimensional description of the finished protein.

The relevant distinction therefore becomes between additive information transmitted explicitly in the molecular message and configurational information implicitly made

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available by structure and reusable physical constraints within a finite space $\mathcal{P}$ that absorbs all environmental effects of execution for all possible configurations.

The paper does not argue that mRNA is an absolutely minimal message, nor does it attempt to calculate how many bits would be required for an alternative spatial encoding.
It argues only that the explicit description of the final coordinates in the composition of the specific configuration is not part of the ordinary genetic message and that the GEDT formalism can represent this level of separation.

5.3 Heme and AHSP: why M_Π should not be assumed null

Heme is synthesized through a distinct pathway and subsequently incorporated into globins. Furthermore, free α-globin is a potentially unstable and reactive species. Alpha-Hemoglobin Stabilizing Protein (AHSP) is an erythroid-specific chaperone that binds free α-globin, increases its stability, limits its reactivity, and participates in the early stages of hemoglobin biosynthesis and assembly.

For GEDT, this point is conceptually useful. If AHSP is treated as a constantly available component of the cellular context, its effect is incorporated into the definition of the feasible space $\mathcal{P}$. If, however, the model separates the physicochemical background from specific dynamic interventions, AHSP may represent a contribution to M_Π and must therefore be explicitly stated as such. In both cases, its existence demonstrates why, in this biological model, H_{res}^{bio} should not be set to zero by definition.

The theory therefore does not require biology to eliminate any additional configurational costs.
It simply requires distinguishing which part of the uncertainty is absorbed by a reusable structure and which part remains to be resolved dynamically.

6. Quasi-identifying regime, equivalence regime and hemoglobin variants

If, conditionally on F, S, in the finite space $\mathcal{P}$, the distribution of Π is strongly concentrated on a single dominant functional configuration, the system can be described as quasi-identifying and H_{res}^{bio} can be small. If multiple conformational states or assemblies remain accessible with non-negligible probability, the system is best represented by an equivalence regime with $H_{res}^{bio} > 0$.

6.1 HbS: sensitivity of a critical position

In hemoglobin S, the substitution historically referred to as β^6 Glu → Val modifies a β-globin surface and, in the deoxygenated state, promotes intermolecular interactions that lead to the polymerization of HbS. A local modification of the primary structure can therefore profoundly alter the supramolecular assembly landscape and the cellular phenotype.

In this work, HbS is not used as "proof" of a mathematically identifying regime.
It is used as an example of high configurational sensitivity: some discrete coordinates of S have much greater weight than others in determining the distribution of accessible global states.

6.2 Hb G-Copenhagen: configurational tolerance

Hb G-Copenhagen is a β-globin variant with an Asp → Asn substitution at position β^{47} in the mature chain numbering (p.Asp^{48}Asn in the numbering including the initiator).
Available clinical and molecular sources do not associate the heterozygous variant with clinically significant symptoms and report normal relative stability.
This variant is a useful example of the opposite case: not every local modification produces a significant macroscopic change in function. In the language of the model, different local realizations of S may fall into the same observable functional class.
Again, this example is illustrative and does not in itself constitute a quantitative estimate of H_{res}^{bio}.

7. Interpretation of the additional transcript M_{Π}

In GEDT, M_{Π} is the additional transcript needed to resolve residual uncertainty about the configuration. In biological mapping, it need not be literally identified with a specific RNA molecule. It is an informational variable of the model.

Depending on the problem, M_{Π} can represent signals, interactions, or dynamic constraints not already included in S or the finite space $\mathcal{P}$, and can assume any molecular or energetic constitution. Conceptual candidates include chaperone assistance, cofactor delivery, subcellular targeting, post-translational modifications, or transient associations with molecular partners.

This flexibility requires discipline in defining the system boundary.
The same factor cannot be simultaneously counted as part of the finite space $\mathcal{P}$ and as M_{Π}.
Any future quantitative application will therefore have to specify ex ante the choice of variable partitioning.

8. Economic-structural corollary: interpretation, not measurement

The GEDT includes a structural-economic corollary that compares the cost of maintaining a reusable structure with the cost of respecifying configurational information at each execution. In the biological domain, an analogy can be drawn between a hypothetical semantically explicit architecture, which would convey detailed configurational instructions, and a structural architecture in which part of this burden is absorbed by sequence and physical constraints.

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Let

$$C_{struct} = C_S + \sum_{t=1}^{Z} \left(C^{S}_{local,t} \right),$$

While, for the explicit architecture:

$$C_{explicit} = C_E + \sum_{t=1}^{Z} \left(C^{E}_{local,t} + C_{add,t} \right).$$

Let

$$\Delta C_t := \left(C^{E}_{local,t} + C_{add,t} \right) - C^{S}_{local,t}$$

where ΔC_t represents the cost difference between the explicit architecture and the structural architecture during execution at cycle t.

Therefore

$$C_s - C_E < \sum_{t=1}^{Z} \Delta C_t. \quad (6)$$

where C_S and C_E represent the cost of the reusable structural architecture and the explicit architecture, respectively; $C_{local,t}^{S}$ and $C_{local,t}^{E}$ are the respective local costs at execution t; $C_{add,t}$ is the additional configurational cost required by the explicit architecture at execution t; and Z is the number of executions considered.
The index E here indicates the “explicit” architecture of the original corollary and not an environmental variable.

Equation (6) should be read as a conceptual condition derived from the original formalism: the advantage emerges when the cost of the reusable structure and its local costs is lower than the cumulative cost of the additional information that a more explicit solution would provide over time.
In this application, C_S, C_E, or ΔC_t are not specifically measured, and therefore a metabolic advantage is not quantitatively demonstrated.

The hypothesis is that the reuse of physicochemical laws and molecular structures can be described as a form of informational amortization. Configurational information is not eliminated: it is distributed between the primary structure, the environment and the execution dynamics.

9. Operational predictions and verification program

To transform mapping into a stronger scientific program, the theory must produce comparable quantities across biological systems. This first application suggests at least four testable directions:

- Comparing proteins that spontaneously refold in vitro with strongly chaperone-dependent proteins: given the same definition of the executive constraints absorbed within the relative finite space $\mathcal{P}$, the second group should show a greater operational residue attributable to M_{Π}.
- Estimate H_{res}^{bio} from experimental or computational conformational ensembles.
- Evaluate the effect of point mutations on the concentration of the configurational distribution: critical mutations should change the distribution of Π more markedly than tolerated conservative substitutions.
- Define a reproducible system boundary and measure how much additional information must be introduced as the finite space $\mathcal{P}$ is progressively depleted of reusable components, for example by removing one chaperone or cofactor at a time.

10. Application Limitations

First, the work proposes a theoretical mapping and not an experimental measure of conformational entropy. The quantities H_{res}^{bio} and D_{epi}^{bio} are formally defined but not numerically estimated in this manuscript.

Second, the partition between S, $\mathcal{P}$, and M_{Π} depends on the model boundary.
This dependence does not invalidate the formalism, but requires explicit declaration of the variables in every quantitative study.
Third, protein folding, used as a model study in this application, is not universally identifiable. Intrinsically disordered proteins, metamorphic proteins, aggregation, misfolding, and multiple states require distributions of Π that are not concentrated on a single outcome.

Fourth, the HbA example does not demonstrate that evolution has minimized a specific cost function of the GEDT. The economic corollary here is interpretative and requires independent measurements to be tested.
Fifth, the work does not propose an alternative to thermodynamics, kinetics, or structural biology.
The GEDT operates at a different descriptive level: it organizes the information burden necessary to determine a global configuration.

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11. Discussion

The potential value of the bio-epistemic application does not consist in stating that "the sequence of the primary structure contains all the information" in an absolute sense.
Such a formulation would be too strong and biologically context-dependent.
The more precise point is that, once a finite constrained space $\mathcal{P}$ is fixed, the primary structure S can greatly reduce the uncertainty about the configuration Π without the final coordinates being explicitly transmitted. This allows the problem to be reinterpreted in terms of information allocation.
A portion is available in the primary structure; a portion resides in physical constraints and reusable environmental conditions; a portion can remain dynamic and require additional components.
The Bio-Epistemic Model proposes the interpretation of these relationships through the application of the GEDT formalism as a study of conditional entropy: $H(\Pi \mid F) = I(\Pi ; S \mid F) + H(\Pi \mid F, S)$.
Hemoglobin is particularly instructive because it simultaneously displays all three levels. The primary sequence directs a significant portion of the architecture; the heme and erythroid context contribute to the physical execution system; AHSP demonstrates that a specific dynamic factor can stabilize an otherwise problematic subunit. Finally, HbS and HbG-Copenhagen show that the informational effect of a local modification depends on position and context, not simply on the counting of substitutions.
From this perspective, folding is not presented as a "proof" of the GEDT.
It is proposed as a natural domain in which the theory generates a coherent and potentially measurable conceptual partition.
The scientific validity of this extension will depend on the ability to transform H_{res}^{bio} and M_{Π} into measurable operational quantities comparable across proteins and experimental conditions.

12. Evolutionary structuralization and imperfect biological compression

From the bio-epistemic perspective developed in this work, evolution may be interpreted as a process capable of progressively redistributing the informational burden required for biological configuration. This interpretation does not assume that natural selection explicitly minimizes an entropy function, nor that evolution acts as a teleological optimization algorithm. It instead proposes that selection can retain structural arrangements that repeatedly constrain biological systems toward viable or functional regions of their admissible configuration space.

Within the present mapping, the ordered primary sequence and its covalent organization constitute the reusable structural component S, whereas the realized functional configuration is represented by Π. The finite set $\mathcal{P}$ represents the admissible configurational space under the declared physicochemical and cellular conditions. Accordingly,

$$H(\Pi \mid F) = I(\Pi ; S \mid F) + H(\Pi \mid F, S)$$

can be read evolutionarily as a partition between configurational uncertainty constrained by persistent structure and uncertainty that remains to be resolved during biological execution.

A sequence variant that more strongly restricts the conditional distribution of Π toward configurations compatible with biological function may therefore reduce the residual term

$H_{res}^{bio} = H(\Pi \mid F, S)$.

Conversely, variants that destabilize recurrent structural relationships, broaden the accessible conformational landscape, or increase dependence on execution-specific assistance may increase the residual configurational burden.

This interpretation does not imply that lower residual entropy is universally favored. Biological systems may require flexibility, conformational multiplicity, regulatory switching, disorder, or environmentally responsive states. The relevant evolutionary quantity is therefore not minimal uncertainty per se, but the compatibility between structural constraint and biological function.

Under this interpretation, evolutionary history can progressively convert repeatedly useful configurational regularities into persistent structural constraints. In informational terms, part of what would otherwise need to be resolved dynamically at each execution may become structurally pre-constrained by inherited organization.

This process may be described as a form of **imperfect biological compression**. The term "compression" is not used here to imply literal data compression or lossless encoding of an organism into its genome. Rather, it indicates that inherited structure can represent, through compact generative constraints, a much larger space of possible phenotypic and configurational outcomes without explicitly specifying the complete microscopic state of each realization.

The compression is necessarily imperfect because biological execution remains conditional on environmental context, stochastic processes, molecular interactions, regulatory networks, cofactors, chaperones, and other dynamic contributions. Persistent genetic or structural information therefore does not replace execution; it constrains the space within which execution occurs.

Adult human hemoglobin provides a simple example of this distinction. The globin sequences do not encode a Cartesian coordinate list of the finished $\alpha_2\beta_2$ tetramer. Nevertheless, they strongly constrain the molecular interactions compatible with the functional architecture. Heme availability, erythroid conditions, AHSP activity, and other cellular factors contribute additional constraints or dynamic assistance depending on the selected model boundary. The resulting configuration is thus generated by the interaction between inherited structural information and recurrent execution conditions rather than by explicit transmission of the complete final geometry.

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12.1 Corollary — Evolutionary reduction of the execution-specific configurational bound

The GEDT reconstruction bound introduced in Section 3 provides a natural way to formalize the preceding interpretation. For a biological system observed under a fixed and explicitly declared model boundary, define the execution-specific configurational lower bound as

$B_{dyn} := \max\{0, H(\Pi \mid F, S) - \delta_{\mathcal{P}}(\epsilon_{\Pi})\}$.

By the GEDT bound,

$E[\ell(M_{\Pi})] \geq B_{dyn}$.

Therefore, B_{dyn} does not represent the complete metabolic, energetic, molecular, or informational cost of folding. It represents only a lower bound on the additional configurational information required once F, S, and the admissible configuration space have been declared.

Consider two evolutionarily related structural realizations, S_t and S_{t+1}, evaluated under the same definition of F, $\mathcal{P}$, reconstruction error ϵ_{Π}, and model boundary.

Define

$B_{dyn}(t) = \max\{0, H(\Pi_t \mid F_t, S_t) - \delta_{\mathcal{P}}(\epsilon_{\Pi})\}$ and $B_{dyn}(t+1) = \max\{0, H(\Pi_{t+1} \mid F_{t+1}, S_{t+1}) - \delta\mathcal{P}(\epsilon_{\Pi})\}$.

We may then define the **evolutionary configurational gain**

$\Gamma_{evo} := B_{dyn}(t) - B_{dyn}(t+1)$.

If

$\Gamma_{evo} > 0$,

the later structural realization requires, at the level of the GEDT lower bound, less execution-specific configurational information than the earlier realization.

If

$\Gamma_{evo} = 0$,

no reduction of this lower bound is detected.

If

$\Gamma_{evo} < 0$,

the later realization retains a greater residual configurational burden under the adopted model.

This quantity must not be interpreted as a universal measure of evolutionary fitness. Selection may favor increased dynamic flexibility, regulatory complexity, environmental responsiveness, or functional multiplicity even when B_{dyn} increases.

The quantity Γ_{evo} therefore measures only a change in configurational informational burden under a fixed formal description.

The corollary yields a potentially testable prediction. For homologous systems performing comparable functions under controlled model boundaries, evolutionary changes that increase the concentration of the conditional distribution of functional configurations should, all else being equal, decrease $H(\Pi \mid F, S)$ and consequently reduce B_{dyn}.

This provides a possible quantitative bridge between structural evolution and the bio-epistemic notion of persistent configurational information.

12.2 Genomic complexity and the C-value problem: a bio-epistemic interpretation

The preceding framework also suggests a cautious interpretation of the well-known absence of a simple proportional relationship between genome size and organismal complexity.

The bio-epistemic model does not propose to solve the C-value problem, nor does it imply that genome size directly measures structural information S. Genome size includes coding, regulatory, repetitive, mobile, and other sequence components whose biological roles and informational relevance are heterogeneous.

Nevertheless, the present framework suggests that organismal complexity need not require an explicit genomic description of every spatial, relational, temporal, and functional configuration realized during development and physiology.

A finite inherited description can participate in the generation of a much larger phenotypic configuration space when it operates together with reusable physical laws, molecular interaction rules, cellular architecture, regulatory feedback, developmental constraints, and environmental conditions.

Thus, the informational relation between genome and phenotype is not necessarily one of direct explicit description. A relatively compact persistent structure can constrain a much larger space of admissible biological configurations.

In GEDT language, biological complexity may therefore depend not only on the explicit size of the inherited description, but also on the efficiency with which persistent structural information constrains the distribution of biologically realizable configurations.

This provides a possible interpretation of why increasing phenotypic or organismal complexity does not require a proportional increase in raw genomic length. The relevant quantity may not be the total number of stored symbols alone, but the configurational consequences generated by their interaction with reusable physical and biological constraints.

This claim remains conceptual. Establishing whether such a relationship explains any measurable component of genome-size variation would require comparative quantitative studies across taxa, explicit definitions of F, S, $\mathcal{P}$, and Π, and empirical estimates of the

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corresponding configurational distributions.

12.3 Mycelial networks as a distributed biological analogue of epistemic dispersion

The bio-epistemic interpretation developed above is not necessarily restricted to protein folding. A second biological domain in which a similar informational separation may emerge is represented by fungal mycelial networks.

Filamentous fungi grow as interconnected networks of hyphae whose architecture continuously adapts to heterogeneous and changing environments. Local hyphal regions interact with nearby resources, while the colony as a whole redistributes nutrients, reinforces or abandons transport pathways, and reorganizes its network in response to damage, competition, and resource availability. Experimental and theoretical studies have therefore described fungal mycelia as adaptive decentralized transport networks in which local processes can affect colony-level behavior across much larger spatial scales.

This organization suggests a natural analogy with epistemic dispersion. The individual local component does not need to contain an explicit representation of the complete colony-wide configuration in order to participate in a coherent global response. Instead, global behavior emerges from the interaction between local sensing, persistent network topology, resource flows, and dynamically propagated constraints.

A possible GEDT mapping can therefore be formulated as follows [13, 14].

Let F_{myc} denote the collection of small, locally isolated hyphal fragments into which the global mycelial network can be decomposed. These fragments represent the local components considered independently of the complete global organization of the network.

Let S_{myc} denote the persistent structural core of the mycelial architecture: the maximal reusable organization that recurs across admissible realizations of the network and constrains how local fragments can be related and recombined.

Let $\mathcal{P}_{myc}$ denote the finite space of admissible mycelial structures compatible with the relevant operational constraints, including physical, biological, environmental, and topological restrictions.

Finally, let

$\Pi_{myc} \in \mathcal{P}_{myc}$

represent one specific admissible global structural configuration of the mycelium.

The bio-epistemic decomposition then becomes

$H(\Pi_{myc} \mid F_{myc}) = I(\Pi_{myc} ; S_{myc} \mid F_{myc}) + H(\Pi_{myc} \mid F_{myc}, S_{myc})$

Under this interpretation, the term

$I(\Pi_{myc} ; S_{myc} \mid F_{myc})$

measures the reduction in uncertainty about the global mycelial configuration produced by knowledge of the persistent structural core, once the local fragments are fixed.

The residual term

$H(\Pi_{myc} \mid F_{myc}, S_{myc})$

represents the remaining uncertainty among the admissible global configurations after both the local fragments and the reusable mycelial structure have been specified.

The role of environmental sensing, nutrient distribution, local chemical conditions, electrical activity [15], mechanical constraints, and other execution-dependent biological factors is therefore not identified with F_{myc}.
Depending on the chosen model boundary, these factors contribute instead to the definition of the admissible configuration space $\mathcal{P}_{myc}$, or to additional execution-specific information required to resolve the realized configuration Π_{myc}.

Using the canonical-core construction introduced in GEDT,
let O denote the operational constraints, let ${\mathcal{P}_0}^{myc}$ be the corresponding admissible mycelial configuration family, and let $\sigma : {\mathcal{P}_0}^{myc} \to \mathcal{S}$ map each admissible configuration to its structural description under the partial order $\preceq$.

Let

${s_0}^{myc} \in \mathcal{S}$

be the canonical core — a distinguished element of the structural-core space $\mathcal{S}$ of the mycelium — determined by the admissible configuration family associated with O.

Whenever the corresponding greatest lower bound exists, GEDT defines the canonical reusable structural core as

$$s_O^{myc} := \bigwedge_{\pi \in \mathcal{P}_O^{myc}} \sigma(\pi)$$

or equivalently,

${s_0}^{myc} = \max_{\preceq} \{s \in \mathcal{S} : s \preceq \sigma(\pi)\ \forall \pi \in {\mathcal{P}_0}^{myc}\}$.

Thus ${s_0}^{myc}$ is the greatest structural object contained in every structural description $\sigma(\pi)$ associated with an admissible configuration $\pi \in {\mathcal{P}_0}^{myc}$: the structural analogue of a greatest common divisor. It contains only relations, invariants, roles, or constraints that survive across the entire admissible configurational family and does not encode the execution-

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specific identity of any single configuration.

13. Conclusion

Protein folding offers a natural case study for General Epistemic Dispersion Theory (GEDT) because the functional three-dimensional configuration is not ordinarily conveyed as an explicit list of spatial coordinates. In the proposed mapping, the identity of the residues constitutes F, the primary sequence S, the physical-cellular environment is absorbed into the finite space of admissible configurations $\mathcal{P}$, and the final structure or ensemble Π.

The quantity $I(\Pi ; S \mid F)$ measures the amount of configurational uncertainty absorbed by the structure, while $H_{res}^{bio} = H(\Pi \mid F, S)$ describes the residue that remains to be resolved.
In quasi-identifying cases, this residue may be small; in systems with multiple states or greater reliance on dynamical assistance, it may remain positive.
Adult hemoglobin shows that this distinction is compatible with known biology: the α and β sequences strongly restrict the conformational space, heme and the environment participate in functional stabilization, AHSP provides an example of additional molecular assistance, and different point variants can have radically different global consequences.

The model does not replace the biochemistry of folding and does not demonstrate reliable quantitative energy savings; instead, it proposes an informational language to distinguish reusable structure, residual uncertainty, and dynamic information.

This distinction constitutes the main theoretical result of the application and defines a program for future quantitative verification.

The same formalism further suggests an evolutionary interpretation in which persistent inherited constraints may reduce part of the execution-specific configurational burden, without implying teleological optimization or a universal tendency toward minimal residual entropy.

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